BIOINF-2018-2693 - View Abstract

> ISoLDE: a data-driven statistical method for the inference of allelic imbalance in datasets with reciprocal crosses

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* Recommendation

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Comments to Associate Editor

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Enter the reasons for your recommendations as to the acceptability or rejection of this paper.

Dear Editor: This manuscript by Christelle Reynès et al. has good originality, decent quality, and solid data support. The method is also implemented as a Bioconductor package. My suggestion is: Accept with minor revisions Best, Zhen Gao, PhD Senior Scientist (Bioinformatics) University of Miami

Comments to the Author

ΩSpecial Characters

Please do not include any statement that will indicate your judgment as to the acceptability of the paper. If possible, submit your comments in the following order:
1) General comments
2)Specific comments for revision: a) major; b) minor.

Dear Authors: The Allelic Imbalance (AI) of the same gene within a single cell is an important and interesting genomic phenomena. The current statistical analysis of AI relies on model-based methods. However, the assumption of the sample distribution and the model selection has significant effects on the performance, accuracy, and confidence of the inference. Minor deviations of the collected data may lead to false discovery or failure to discover meaningful results. The authors designed a non-parametric statistic framework to learn the sample distribution from the data themselves and avoid the prior assumption of distribution. When there are only two replicates in the reciprocal crosses, predefined thresholds are adopted by this method. Through multiple benchwork comparisons, the ISoLDE method was shown quite robust and relative conservative. The organization of this manuscript is logical, the written is clear and easy to understand. In addition, this ISoLDE method has been implemented as a Bioconductor package for adopted by other researches. This research is innovative and produced good and meaningful results. The following are the flaws, possible mistakes, or typo etc., which need to be corrected or clarified. 1) The authors used sensitivity and specificity data quite a lot, that's good. But I strongly suggest the authors add the F1 score to combine the sensitivity and specificity in all situations. As the F1 score could provide a direct measure of performance when the performances in sensitivity and specificity are discordant. 2) In line 50 of page 2: I believe the formula has an error, as the left and right parts of the denominator are not balanced. (ngi,j,M) / (ngi,j,P) vs (ngi,j,P) / (ngi,j,P) , note the M and P 3) In line 10 of page 4: the lower the variability an 'AI' call. I guess to it could be changed to 4) All plots should be vector graphs, such as SVG format, as bitmap graph results in very poor quality, I cannot see details in Figures 1, 3 and quite a few supplementary Figures. 5) In line 50-60 of page 5: Authors used repeatedly "respectively" in this paragraph, it's a bit confusing. 6) In Figure 2 in Page 5: "AI-AI" and "AI-BA" might need to explain, such as "the AI-AI pair in legend" 7) Authors used repeatly "all in all", this is not an academic style, I suggest it could be changed to some other expression, such as "in summary", "in short", "in conclusion" 8) In the Supplementary_Table_S1.xlsx, the line 6 (start with Crowley) have mistakes: the scale is "cf. Zou", and all data after the column "modelling" are missing. 9) In the "Supplementary_methods_figures.docx", the section -- Simulation details -- Generating read numbers, seqdepth = 5.10^6, it looks like a typo. The depth is impossible to be that high. In the section--Mapping Methods, "Mapping was performed using regular align software. It's not enough, the software name and version should be clearly demonstrated. 10) The perl scripts mentioned in the "Supplementary_methods_figures.docx" might need to be supplied for reviewer and researcher.